

LECTURE: SEPARATION OF VARIABLES (II)

Today: Separation of variables but with the Wave Equation

1. SEPARATION OF VARIABLES

Video: Separation of Variables (Wave)

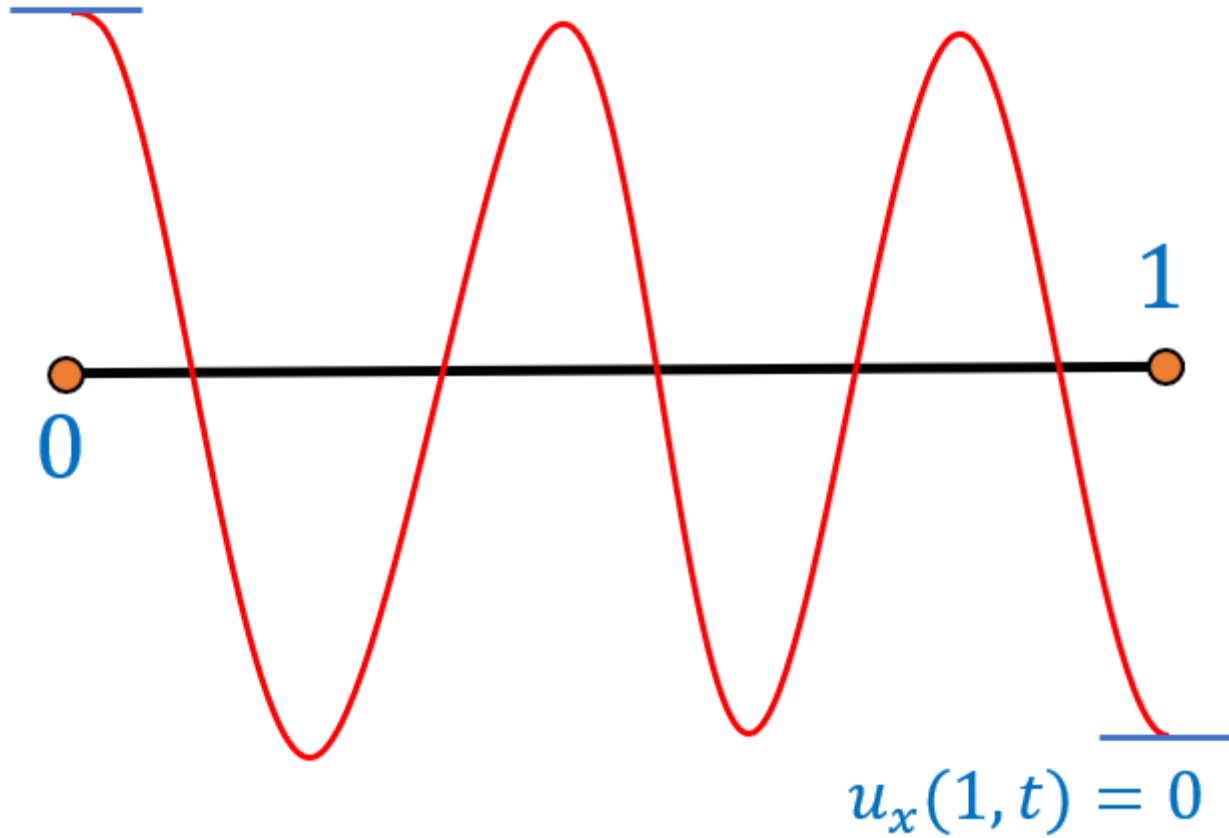
Note: The video covers the same problem as below, except it deals with the case $0 < x < \pi$ and with a different initial condition

Example 1:

$$\left\{ \begin{array}{l} u_{tt} = c^2 u_{xx} \\ u_x(0, t) = 0 \\ u_x(1, t) = 0 \\ u(x, 0) = x^3 \\ u_t(x, 0) = \cos(3\pi x) \end{array} \right.$$

This is the wave equation, but this time we're assuming the *velocity* at the endpoints is 0

$$u_x(0, t) = 0$$



PART I: SEPARATION OF VARIABLES

STEP 1:

Suppose $u(x, t) = X(x)T(t)$

STEP 2: Plug this into the PDE

$$\begin{aligned}
u_{tt} &= c^2 u_{xx} \\
(X(x)T(t))_{tt} &= c^2 (X(x)T(t))_{xx} \\
X(x) T''(t) &= c^2 X''(x) T(t)
\end{aligned}$$

STEP 3: Put the X terms on one side and the T terms on the other making sure any constants go with T

$$\frac{T''(t)}{c^2 T(t)} = \frac{X''(x)}{X(x)} = \lambda \text{ (See last time)}$$

$$\begin{aligned}
\frac{X''(x)}{X(x)} &= \lambda \Rightarrow X''(x) = \lambda X(x) \\
\frac{T''(t)}{c^2 T(t)} &= \lambda \Rightarrow T''(t) = \lambda c^2 T(t)
\end{aligned}$$

PART II: $X(x)$ equation

So far: $X''(x) = \lambda X(x)$

NEW Boundary Conditions

$$\begin{aligned}
u(x, t) &= X(x)T(t) \\
u_x(x, t) &= (X(x)T(t))_x = X'(x)T(t)
\end{aligned}$$

$$\begin{aligned}
u_x(0, t) = 0 &\Rightarrow X'(0)\cancel{T(t)} = 0 \Rightarrow X'(0) = 0 \\
u_x(1, t) = 0 &\Rightarrow X'(1)\cancel{T(t)} = 0 \Rightarrow X'(1) = 0
\end{aligned}$$

Hence we get the ODE

$$\begin{cases} X''(x) = \lambda X(x) \\ X'(0) = 0 \\ X'(1) = 0 \end{cases}$$

PART III: BOUNDARY-VALUE PROBLEM

Auxiliary Equation: $r^2 = \lambda$

Case 1: $\lambda > 0$

Then $r^2 = \lambda = \omega^2 \Rightarrow r = \pm\omega$ in which case we get

$$X(x) = Ae^{\omega x} + Be^{-\omega x}$$

$$X'(x) = A\omega e^{\omega x} - B\omega e^{-\omega x}$$

$$X'(0) = A\omega - B\omega = 0 \Rightarrow A\cancel{\omega} = B\cancel{\omega} \Rightarrow A = B$$

$$X(x) = Ae^{\omega x} + Ae^{-\omega x}$$

$$X'(1) = 0$$

$$A\omega e^{\omega} - A\omega e^{-\omega} = 0$$

$$\cancel{A}\omega e^{\omega} = \cancel{A}\omega e^{-\omega}$$

$$e^{\omega} = e^{-\omega}$$

$$\omega = -\omega$$

$$\omega = 0$$

But then $\lambda = \omega^2 = 0^2 = 0$ which contradicts $\lambda > 0 \Rightarrow \Leftarrow$

Conclusion: In this case, we have no nonzero solutions

Case 2: $\lambda = 0$

Aux: $r^2 = 0 \Rightarrow r = 0$ (repeated twice)

$$X(x) = A + Bx$$

$$X'(x) = B$$

$$X'(0) = 0 \Rightarrow B = 0$$

$$X(x) = A + 0x = A = A \times 1$$

But then in this case $X'(1) = 0$ is automatically satisfied.

Conclusion: $\lambda = 0$ is an eigenvalue with eigenfunction $X(x) = 1$

Case 3: $\lambda < 0$

In this case $\lambda = -\omega^2$ where $\omega > 0$

Aux: $r^2 = \lambda = -\omega^2 \Rightarrow r = \pm\omega i$

$$X(x) = A \cos(\omega x) + B \sin(\omega x)$$

$$X'(x) = -A\omega \sin(\omega x) + B\omega \cos(\omega x)$$

$$X'(0) = -A\omega \sin(0) + B\omega \cos(0) = B\omega = 0 \Rightarrow B = 0$$

$$X(x) = A \cos(\omega x)$$

$$X'(1) = 0$$

$$\cancel{-A\omega} \sin(\omega) = 0$$

$$\sin(\omega) = 0$$

$$\omega = \pi m \quad m = 1, 2, \dots$$

Here again m starts at 1 since we have $\omega > 0$

In that case $\lambda = -\omega^2 = -(\pi m)^2$ and $X(x) = \cos(\omega x) = \cos(\pi m x)$

Amazing Coincidence: If you let $m = 0$ then $\lambda = -(\pi 0)^2 = 0$ and

$$X(x) = \cos(\pi 0 x) = \cos(0) = 1$$

Which *includes* the previous step!

Conclusion:

Eigenvalues: $\lambda = -(\pi m)^2$ ($m = 0, 1, 2, \dots$)

Eigenfunctions: $X(x) = \cos(\pi m x)$ ($m = 0, 1, 2, \dots$)

PART IV: T EQUATION

$$\begin{aligned}\frac{T''(t)}{c^2 T(t)} &= \lambda = -(\pi m)^2 \\ T''(t) &= -(\pi m)^2 c^2 T(t) \\ T''(t) &= -(\pi m c)^2 T(t)\end{aligned}$$

Special case: If $m = 0$ we get

$$T''(t) = 0 \Rightarrow T(t) = A_0 + B_0 t \text{ for some } A_0 \text{ and } B_0$$

General case: If $m = 1, 2, \dots$ the auxiliary equation becomes

$$\mathbf{Aux:} \quad r^2 = -(\pi m c)^2 \Rightarrow r = \pm \pi m c i$$

$$\text{Hence } T(t) = A_m \cos(\pi m c t) + B_m \sin(\pi m c t)$$

Conclusion:

Special Case: For $m = 0$ the following solves our PDE

$$u(x, t) = X(x)T(t) = (A_0 + B_0 t) \cos(\pi 0 x) = A_0 + B_0 t$$

General Case: For $m = 1, 2, \dots$ the following solves our PDE

$$u(x, t) = X(x)T(t) = (A_m \cos(\pi m c t) + B_m \sin(\pi m c t)) \cos(\pi m x)$$

PART V: LINEARITY

Taking linear combinations, we get

$$\begin{aligned} u(x, t) &= \text{Special Case} + \text{General Case} \\ &= A_0 + B_0 t + \sum_{m=1}^{\infty} (A_m \cos(\pi m c t) + B_m \sin(\pi m c t)) \cos(\pi m x) \end{aligned}$$

PART VI: INITIAL CONDITION

$$u(x, 0) = A_0 + B_0(0) + \sum_{m=1}^{\infty} \left(A_m \underbrace{\cos(\pi m c 0)}_1 + B_m \underbrace{\sin(\pi m c 0)}_0 \right) \cos(\pi m x)$$

$$x^3 = A_0 + \sum_{m=1}^{\infty} A_m \cos(\pi m x)$$

$$x^3 = A_0 \cos(\pi 0 x) + \sum_{m=1}^{\infty} A_m \cos(\pi m x)$$

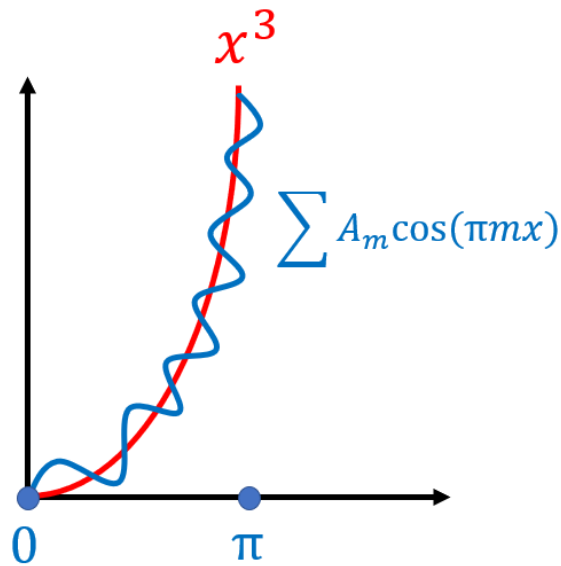
$$x^3 = \sum_{m=0}^{\infty} A_m \cos(\pi m x)$$

This time we have a cos problem!

Fourier Question:

Can you find A_m such that

$$x^3 = \sum_{m=0}^{\infty} A_m \cos(\pi m x)$$

**PART VII: INITIAL VELOCITY**

Important Remark: Since $u_t(x, 0)$ is already written in terms of $\cos(\pi m x)$ we don't need Fourier series here!

$$\begin{aligned}
u_t(x, t) &= (A_0 + B_0 t)_t + \sum_{m=1}^{\infty} (A_m \cos(\pi m c t) + B_m \sin(\pi m c t))_t \cos(\pi m x) \\
&= B_0 + \sum_{m=1}^{\infty} ((-A_m \pi m c) \sin(\pi m c t) + (B_m \pi m c) \cos(\pi m c t)) \cos(\pi m x) \\
u_t(x, 0) &= B_0 + \sum_{m=1}^{\infty} \left((-A_m \pi m c) \underbrace{\sin(\pi m c 0)}_0 + (B_m \pi m c) \underbrace{\cos(\pi m c 0)}_1 \right) \cos(\pi m x) \\
\cos(3\pi x) &= B_0 + \sum_{m=1}^{\infty} (\pi m c B_m) \cos(\pi m x)
\end{aligned}$$

Using the definition of a sum we get

$$\begin{aligned}
& B_0 + (1\pi c B_1) \cos(\pi x) + (2\pi c B_2) \cos(2\pi x) + (3\pi c B_3) \cos(3\pi x) + (4\pi c B_4) \cos(4\pi x) \\
&= \\
&= 0 + 0 \cos(\pi x) + 0 \cos(2\pi x) + 1 \cos(3\pi x) + 0 \cos(4\pi x)
\end{aligned}$$

Which gives

$$B_0 = 0 \text{ and } B_1 = 0 \text{ and } B_2 = 0 \text{ and } B_3 = \frac{1}{3\pi c} \text{ and } B_4 = 0 \dots$$

So we are essentially done once we found the A_m

2. INHOMOGENEOUS PROBLEMS

The cool thing is that we can also treat nonzero boundary conditions with a simple subtraction trick.

Example 2:

$$\begin{cases} u_{tt} = c^2 u_{xx} \\ u(0, t) = 7 \\ u(1, t) = 7 \\ u(x, 0) = x^3 \\ u_t(x, 0) = \cos(3\pi x) \end{cases}$$

STEP 1: Trick:

$$\text{Let } v(x, t) = u(x, t) - 7$$

$$\text{Then } v_{tt} = (u - 7)_{tt} = u_{tt} = c^2 u_{xx} = c^2 (v + 7)_{xx} = c^2 v_{xx}$$

$$v(0, t) = u(0, t) - 7 = 7 - 7 = 0$$

$$v(1, t) = u(1, t) - 7 = 7 - 7 = 0$$

$$v(x, 0) = u(x, 0) - 7 = x^3 - 7$$

$$v_t(x, t) = (u(x, t) - 7)_t = u_t(x, t)$$

$$v_t(x, 0) = u_t(x, 0) = \cos(3\pi x)$$

STEP 2: Solve

$$\begin{cases} v_{tt} = c^2 v_{xx} \\ v(0, t) = 0 \\ v(1, t) = 0 \\ v(x, 0) = x^3 - 7 \\ v_t(x, 0) = \cos(3\pi x) \end{cases}$$

Solve for $v(x, t)$ using separation variables

STEP 3: Finally, use

$$u(x, t) = v(x, t) + 7$$

Example 3:

$$\begin{cases} u_{tt} = c^2 u_{xx} \\ u_x(0, t) = 7 \\ u_x(1, t) = 7 \\ u(x, 0) = x^3 \\ u_t(x, 0) = \cos(3\pi x) \end{cases}$$

STEP 1: Trick: Subtract a function whose x -derivative is 7

$$\text{Let } v(x, t) = u(x, t) - 7x$$

$$\text{Then } v_{tt} = (u - 7x)_{tt} = u_{tt} = c^2 u_{xx} = c^2 (v + 7x)_{xx} = c^2 v_{xx}$$

$$v_x(0, t) = u_x(0, t) - 7 = 7 - 7 = 0$$

$$v_x(1, t) = u_x(1, t) - 7 = 7 - 7 = 0$$

$$v(x, 0) = u(x, 0) - 7x = x^3 - 7x$$

$$v_t(x, t) = (u(x, t) - 7x)_t = u_t(x, t)$$

$$v_t(x, 0) = u_t(x, 0) = \cos(3\pi x)$$

STEP 2: Solve

$$\begin{cases} v_{tt} = c^2 v_{xx} \\ v_x(0, t) = 0 \\ v_x(1, t) = 0 \\ v(x, 0) = x^3 - 7x \\ v_t(x, 0) = \cos(3\pi x) \end{cases}$$

Solve for $v(x, t)$ using separation variables

STEP 3: Finally, use

$$u(x, t) = v(x, t) + 7x$$

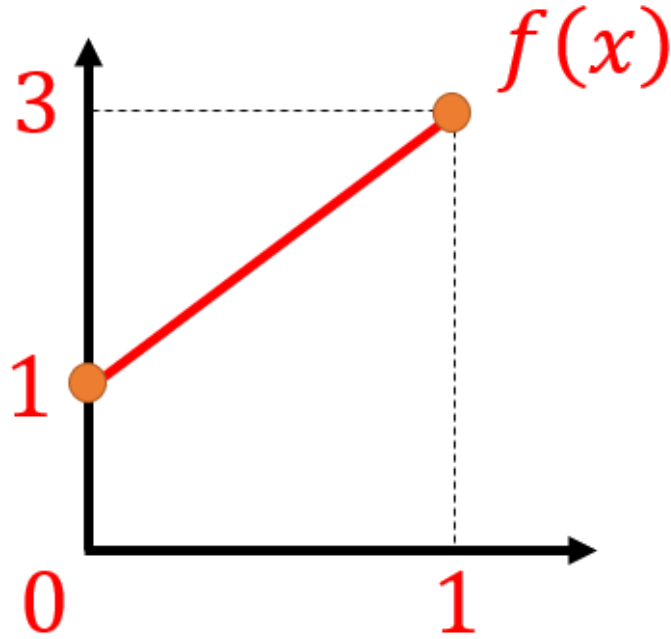
Example 4:

$$\begin{cases} u_{tt} = x^2 u_{xx} \\ u(0, t) = 1 \\ u(1, t) = 3 \\ u(x, 0) = x^3 \\ u_t(x, 0) = \cos(3\pi x) \end{cases}$$

STEP 1: Trick:

$$\text{Let } v(x, t) = u(x, t) - f(x)$$

Where f is a linear function such that $f(0) = 1$ and $f(1) = 3$



$$f(x) = \left(\frac{3-1}{1-0} \right) (x-0) + 1 = 2x + 1$$

$$\text{Let } v(x, t) = u(x, t) - 2x - 1$$

$$\text{Then } v_{tt} = (u - 2x - 1)_{tt} = u_{tt} = c^2 u_{xx} = c^2 (v + (2x + 1))_{xx} = c^2 v_{xx}$$

$$v(0, t) = u(0, t) - 2(0) - 1 = 1 - 1 = 0$$

$$v(1, t) = u(1, t) - 2(1) - 1 = 3 - 3 = 0$$

$$v(x, 0) = u(x, 0) - 2x - 1 = x^3 - 2x - 1$$

$$v_t(x, t) = (u(x, t) - 2x - 1)_t = u_t(x, t)$$

$$v_t(x, 0) = u_t(x, 0) = \cos(3\pi x)$$

STEP 2: Solve

$$\begin{cases} v_{tt} = c^2 v_{xx} \\ v(0, t) = 0 \\ v(1, t) = 0 \\ v(x, 0) = x^3 - 2x - 1 \\ v_t(x, 0) = \cos(3\pi x) \end{cases}$$

Solve for $v(x, t)$ using separation of variables

STEP 3:

$$u(x, t) = v(x, t) + f(x) = v(x, t) + 2x + 1$$

Aside: We could *in theory* also be able to solve the case where

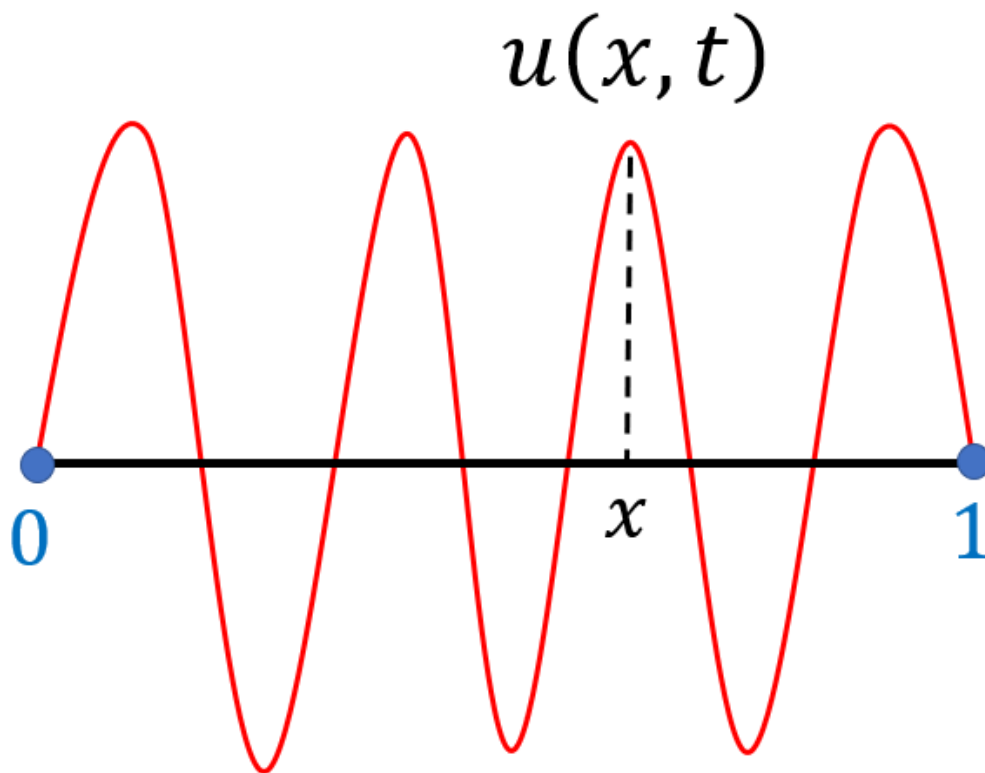
$$u_x(0, t) = 1 \text{ and } u_x(1, t) = 3$$

By subtracting a function whose *derivative* is $2x + 1$ except you would get an *inhomogeneous* wave equation for v

3. EASIER PROBLEM

Example 5: (more practice)

$$\begin{cases} u_{tt} = c^2 u_{xx} \\ u(0, t) = 0 \\ u(1, t) = 0 \\ u(x, 0) = \sin(2\pi x) + 3 \sin(3\pi x) \\ u_t(x, 0) = 4 \sin(2\pi x) \end{cases}$$



PARTS I–V: SEPARATION OF VARIABLES

Similar to the above, using separation of variables, can show

$$u(x, t) = \sum_{m=1}^{\infty} (A_m \cos(\pi m c t) + B_m \sin(\pi m c t)) \sin(\pi m x)$$

PART VI: INITIAL POSITION

This time we can solve for all the constants

$$\begin{aligned}
 u(x, 0) &= \sum_{m=1}^{\infty} \left(A_m \underbrace{\cos(\pi m c 0)}_1 + B_m \underbrace{\sin(\pi m c 0)}_0 \right) \sin(\pi m x) \\
 &= \sum_{m=1}^{\infty} A_m \sin(\pi m x) \\
 &= A_1 \sin(\pi x) + A_2 \sin(2\pi x) + A_3 \sin(3\pi x) + \dots \\
 &= \sin(2\pi x) + 3 \sin(3\pi x) \quad (\text{Initial Condition}) \\
 &= 0 \sin(\pi x) + 1 \sin(2\pi x) + 3 \sin(3\pi x) + \dots
 \end{aligned}$$

$$A_1 = 0 \text{ and } A_2 = 1 \text{ and } A_3 = 3 \text{ and } A_4 = 0 \dots$$

PART VII: INITIAL VELOCITY

$$\begin{aligned}
 u_t(x, t) &= \sum_{m=1}^{\infty} (A_m \cos(\pi m c t) + B_m \sin(\pi m c t))_t \sin(\pi m x) \\
 &= \sum_{m=1}^{\infty} [(-\pi m c A_m) \sin(\pi m c t) + (\pi m c B_m) \cos(\pi m c t)] \sin(\pi m x) \\
 u_t(x, 0) &= \sum_{m=1}^{\infty} \left[(-\pi m c A_m) \underbrace{\sin(\pi m c 0)}_0 + (\pi m c B_m) \underbrace{\cos(\pi m c 0)}_1 \right] \sin(\pi m x) \\
 &= \sum_{m=1}^{\infty} (\pi m c B_m) \sin(\pi m x) \\
 &= (\pi c B_1) \sin(\pi x) + (2\pi c B_2) \sin(2\pi x) + (3\pi c B_3) \sin(3\pi x) + \dots \\
 &= 4 \sin(2\pi x) \quad (\text{Initial Velocity}) \\
 &= 0 \sin(\pi x) + 4 \sin(2\pi x) + 0 \sin(3\pi x) + \dots
 \end{aligned}$$

$$\pi c B_1 = 0 \text{ and } 2\pi c B_2 = 4 \text{ and } \pi 3c B_3 = 0 \text{ and } \dots$$

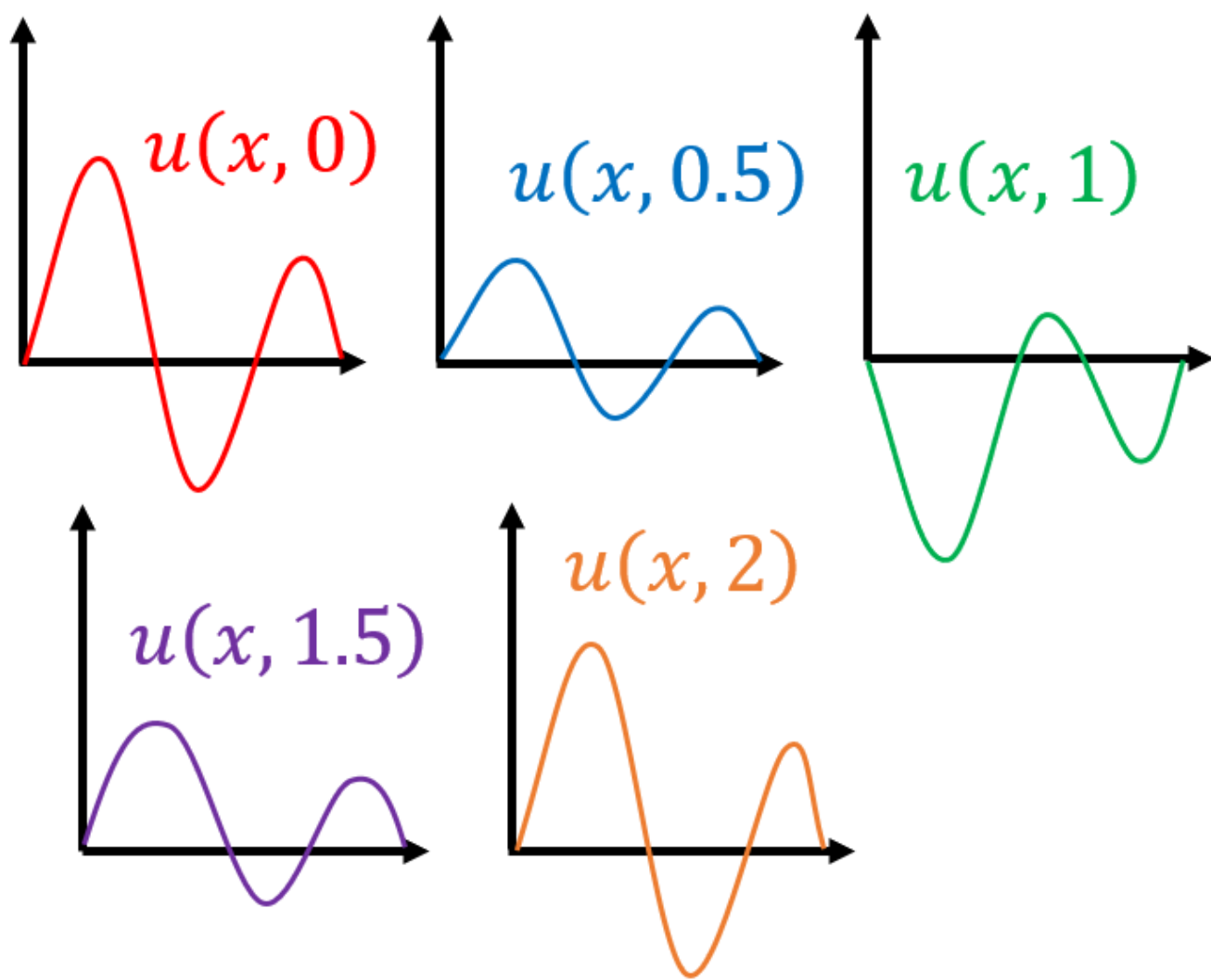
$$B_1 = 0 \text{ and } B_2 = \frac{2}{\pi c} \text{ and } B_3 = 0 \dots$$

Solution: Notice there are no more sums because most terms are 0

$$\begin{aligned} u(x, t) &= \sum_{m=1}^{\infty} (A_m \cos(\pi m c t) + B_m \sin(\pi m c t)) \sin(\pi m x) \\ &= (A_2 \cos(2\pi c t) + B_2 \sin(2\pi c t)) \sin(2\pi x) \\ &\quad + (A_3 \cos(3\pi c t) + B_3 \sin(3\pi c t)) \sin(3\pi x) \\ &= \left(\cos(2\pi c t) + \left(\frac{2}{\pi c} \right) \sin(2\pi c t) \right) \sin(2\pi x) + (3 \cos(3\pi c t)) \sin(3\pi x) \end{aligned}$$

Interpretation:

Initially, at $t = 0$, the solution starts out as $3 \sin(2\pi x) + \sin(3\pi x)$ but oscillates back and forth



Demo: Wave Equation Demo